

Yes. The best way to combine ideas **1 + 2 + 4** is to build the paper around a single stronger question:

When do geopolitical shocks become systemic financial stress events?

This lets you use the existing geopolitical shock data, but the paper's true contribution becomes **financial stability and volatility transmission**, not shock construction. The market-implied stress detection from idea 4 becomes a second layer that tells us whether a geopolitical shock was actually "absorbed" by markets or converted into systemic stress.

A possible title:

When Geopolitical Shocks Become Systemic: Volatility Connectedness, Market-Implied Stress, and Financial Stability in European Equity Markets

or more compact:

Geopolitical Shocks and Financial Stability in European Equity Markets: A Dual News–Market Approach

1. Core conceptual framework

Think of the paper as having three layers.

Layer 1 — News-based geopolitical shock trigger

This is the existing shock measure. But we no longer make it the contribution. We treat it as an external trigger.

Examples:

G_t = aggregate geopolitical shock intensity across countries

B_t = number of countries experiencing shocks on day t

$MaxShock_t$ = maximum country-level shock intensity on day t

The important change is that we avoid saying:

"We propose a new geopolitical shock measure."

Instead, we say:

"We use a validated high-frequency geopolitical conflict shock series as a trigger to study systemic financial-market responses."

That moves the paper away from BIR.

Layer 2 — Financial stability / systemic fragility metrics

This is the main new object.

Using the 19 European equity indices, construct daily indicators of systemic financial stress:

A. European volatility stress

Average absolute return:

$$\text{VolStress}_t = \text{average}_i |r_{\{i,t\}}|$$

B. Tail co-exceedance breadth

Number of markets simultaneously in their volatility tail:

$$\text{TailVolBreadth}_t = \text{number of countries with } |r_{\{i,t\}}| > Q_{\{i,95\}}$$

Or downside distress:

$$\text{TailLossBreadth}_t = \text{number of countries with } r_{\{i,t\}} < Q_{\{i,5\}}$$

C. Average cross-market correlation

Rolling average pairwise correlation across country returns.

D. Volatility connectedness index

Use Diebold–Yilmaz or TVP-VAR connectedness on volatility proxies.

$$\text{TCI}_t = \text{total volatility connectedness index}$$

E. Composite European Market Fragility Index

Combine standardized components:

$$\text{EMFI}_t = \text{PCA}_1(\text{VolStress}_t, \text{TailVolBreadth}_t, \text{AvgCorr}_t, \text{TCI}_t)$$

This becomes a major methodological contribution: a **daily market-based financial stability index** for European equity markets.

Layer 3 — Market-implied stress detector

This is the idea from point 4. But it should be used carefully.

The detector should not replace the geopolitical shock measure. Instead, it answers:

Did the market itself classify the period around a geopolitical shock as systemic stress?

This is powerful because not every geopolitical event matters financially. Some events are geopolitical but not systemic. Others become systemic because they trigger common volatility, tail co-movement, or connectedness.

Possible methods:

Conservative option: PCA anomaly score

Compute the first principal component of volatility/correlation/tail-stress indicators. Define market stress days as top 5% or 1% of the anomaly score.

Stronger option: Hidden Markov Model

Estimate a two- or three-state HMM on market features:

State 1 = calm

State 2 = elevated volatility

State 3 = systemic stress

Then obtain:

$P(\text{SystemicStress}_t)$

More advanced option: graph anomaly detection

Build rolling correlation or volatility networks and detect days when the network topology changes sharply.

For a finance journal, I would start with **HMM + PCA anomaly score as robustness**. It is understandable, publishable, and not too “black-box.”

2. Unified research question

The full research question could be:

Do geopolitical shocks increase systemic fragility in European equity markets, and are financially relevant geopolitical shocks distinguishable by their effect on volatility connectedness and market-implied stress regimes?

This combines all three ideas:

- idea 1: volatility connectedness;
- idea 2: financial stability / fragility index;
- idea 4: market-implied stress detection.

A more elegant formulation:

When does geopolitical news become systemic financial stress?

That is a very strong framing.

3. Main hypotheses

H1 — Geopolitical shocks increase systemic market fragility

After geopolitical shocks, the European Market Fragility Index rises.

$$\text{EMFI}_{t+k} = \alpha + \beta_k G_t + \text{error}_{t+k}$$

Expected result:

$$\beta_k > 0$$

especially for short horizons.

H2 — Geopolitical shocks increase volatility connectedness

The key claim is not simply that volatility rises, but that volatility becomes more interconnected.

$$\text{TCI}_{t+k} = \alpha + \beta_k G_t + \text{error}_{t+k}$$

Expected result:

$$\beta_k > 0$$

This is the spillover/contagion channel.

H3 — Geopolitical shocks are more destabilizing when markets are already fragile

This is very important and could make the paper much stronger.

Use pre-shock connectedness or pre-shock fragility as a state variable:

$$\text{HighFragility}_{t-1} = 1 \text{ if } \text{EMFI}_{t-1} \text{ is above its 75th percentile}$$

Then estimate:

$$\text{EMFI}_{t+k} =$$

α

$+ \beta_k G_t$

$+ \theta_k G_t \times \text{HighFragility}_{t-1}$

$+ \text{error}_{t+k}$

Interpretation:

- β_k = effect of geopolitical shocks in normal markets;
- θ_k = additional effect when the system is already fragile.

This gives the paper a strong financial-stability message:

Geopolitical shocks become systemic mainly when they hit an already-connected or fragile market structure.

H4 — Only some geopolitical shocks become market-implied systemic stress events

Use the HMM or anomaly detector to classify market stress.

Define:

$\text{MarketStress}_t = 1$ if $P(\text{SystemicStress}_t) > 0.8$

Then test whether geopolitical shocks increase the probability of entering market stress:

$\Pr(\text{MarketStress}_{\{t:t+5\}} = 1)$

=

$f(G_t, B_t, \text{PreFragility}_t, G_t \times \text{PreFragility}_t)$

This creates a powerful typology:

Type	Interpretation
News-only geopolitical shocks	Geopolitical events that markets absorb
Market-only stress events	Financial stress not clearly linked to geopolitical shocks
Joint geopolitical-market stress events	Geopolitical shocks that become systemic
Persistent systemic events	Shocks followed by prolonged fragility

The “joint” events are the most interesting.

4. The paper’s main contribution

The contribution would no longer be:

“We construct a geopolitical shock measure.”

It becomes:

“We show when and how geopolitical shocks become systemic financial stress events by combining news-based geopolitical triggers with market-implied fragility, volatility connectedness, and stress-regime detection.”

That is much further away from the BIR paper.

The BIR paper says:

“Here is a validated shock measure.”

The new paper says:

“Using that shock measure, we study whether geopolitical events destabilize the financial system through volatility connectedness and market-implied stress regimes.”

That is a much cleaner separation.

5. Proposed empirical design

Step 1 — Construct daily market fragility indicators

Using the 19 equity-market return series:

1. average absolute return;
2. average squared return;
3. tail volatility breadth;
4. downside tail co-exceedance;
5. rolling average correlation;
6. volatility connectedness index;
7. composite European Market Fragility Index.

The paper should not rely only on one index. The composite index can be the main metric, while individual components are shown separately.

Step 2 — Estimate volatility connectedness

Use one main connectedness framework.

I would recommend one of these:

Option A: Diebold–Yilmaz rolling VAR connectedness

Input:

$$v_{\{i,t\}} = |r_{\{i,t\}}|$$

Estimate rolling VAR and compute forecast-error variance decomposition.

Outputs:

- total connectedness index;
- directional connectedness from country i to country j ;
- net transmitter / receiver position;
- country-level spillover centrality.

Option B: TVP-VAR connectedness

This is more elegant because connectedness evolves smoothly over time. It is also more modern, but slightly harder to implement.

Option C: Correlation-network connectedness

Simpler and more robust:

- rolling correlation matrix;
- network density;
- average correlation;
- eigenvector centrality;
- minimum spanning tree length;
- graph concentration.

For publication, I would use:

TVP-VAR or rolling Diebold–Yilmaz as the main connectedness measure, with rolling correlation networks as robustness.

Step 3 — Construct market-implied stress regimes

Use an HMM on a small vector of systemic indicators:

$X_t = [\text{VolStress}_t, \text{TailVolBreadth}_t, \text{AvgCorr}_t, \text{TCI}_t]$

Estimate 2 or 3 regimes.

A clean three-state model:

State Meaning

- 1 Normal market
- 2 Elevated volatility
- 3 Systemic stress

Then extract:

P_{stress_t} = probability of systemic stress

This becomes the market-implied stress score.

Important: this detector must be estimated **without using the geopolitical shock dates**. Otherwise, the exercise becomes circular.

Step 4 — Test dynamic effects of geopolitical shocks on systemic fragility

Estimate local projections:

$$Y_{t+k} = \alpha + \beta_k G_t + \text{controls} + \text{error}_{t+k}$$

where Y can be:

- European Market Fragility Index;
- total volatility connectedness;
- tail co-exceedance breadth;
- market stress probability;
- average correlation.

Possible controls:

- lagged dependent variable;
- global market return;
- VIX if you later add it;
- day-of-week effects;
- COVID / Ukraine-war dummy if necessary;
- month fixed effects.

Even without many controls, the design can be credible if the main objective is dynamic association rather than strict causality.

Step 5 — State-dependent local projections

This is one of the best methodological additions.

Estimate whether the effect depends on pre-existing fragility:

$$Y_{t+k}$$

=

α

+ $\beta_k G_t$

+ $\theta_k G_t \times \text{HighConnectedness}_{t-1}$

+ $\phi \text{HighConnectedness}_{t-1}$

+ error_{t+k}

This gives a very publishable finding if it works:

Geopolitical shocks have limited systemic effects in calm markets, but they strongly increase fragility when pre-shock connectedness is already elevated.

This is much more interesting than the current result that volatility rises at $k = 4$ and $k = 5$.

Step 6 — Identify systemic geopolitical events

Classify each geopolitical shock event into one of four categories:

Category	Definition
Absorbed geopolitical shock	news shock, no market stress response
Localized financial response	news shock, volatility rises in few markets
Systemic geopolitical stress	news shock followed by high EMFI / high TCI / HMM stress state
Market stress without geopolitical trigger	market stress event with no nearby geopolitical shock

Then compare:

- average volatility response;
- connectedness response;
- tail co-exceedance response;
- duration of stress;
- countries that become transmitters/receivers.

This classification is where idea 4 becomes very useful.

6. Main figures and tables

This paper could have very compelling visuals.

Figure 1 — Timeline of geopolitical shocks and market fragility

Plot:

- European Market Fragility Index;
- shaded market-implied stress regimes;

- vertical lines for major geopolitical shocks.

This immediately shows whether the two layers align.

Figure 2 — Dynamic response of systemic fragility to geopolitical shocks

Local projection response of:

- EMFI;
- TCI;
- tail co-exceedance breadth.

Figure 3 — State-dependent responses

Two lines:

- effect of geopolitical shocks during normal pre-shock markets;
- effect during high-fragility pre-shock markets.

This could be the paper's most important figure.

Figure 4 — Volatility connectedness network before and after major geopolitical events

For example:

- Russia invasion of Ukraine;
- Hamas-Israel shock;
- Iran-Israel escalation;
- major European terrorist events.

Show how network density / transmitter countries change.

Figure 5 — Market-implied stress classification

A matrix or event table:

Event News shock intensity EMFI response TCI response HMM stress probability Classification

This would make the paper concrete and interpretable.

7. Possible final paper structure

Title

When Geopolitical Shocks Become Systemic: Volatility Connectedness and Market-Implied Stress in European Equity Markets

Abstract idea

The paper studies whether geopolitical shocks become systemic financial stress events in European equity markets. Using daily data for 19 equity markets from 2017–2025, the paper combines a validated high-frequency geopolitical shock series with market-based measures of financial fragility, volatility connectedness, and market-implied stress regimes. The key result would be that geopolitical shocks increase systemic fragility primarily when they amplify volatility connectedness or occur during already fragile market conditions.

Sections

1. Introduction
2. Data and geopolitical shock triggers
3. Market-based financial stability and connectedness measures
4. Market-implied systemic stress detection
5. Geopolitical shocks and dynamic fragility responses
6. State dependence and systemic event classification
7. Robustness
8. Conclusion

8. Why this is much better than the current rejected paper

The current rejected paper says:

Geopolitical shocks increase equity-market volatility.

The new paper says:

Geopolitical shocks become financially systemic only when they trigger connectedness, tail co-movement, and market-implied stress regimes.

That is a much richer and more publishable claim.

It also avoids the BIR overlap because:

- BIR = shock construction and validation;
- new paper = financial stability and systemic transmission;
- market-implied stress detector = independent market-based layer;

- connectedness and fragility metrics = new contribution;
 - classification of systemic vs absorbed geopolitical shocks = new research question.
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9. My recommended version

I would not make market-implied detection the main pillar at first. I would use it as an **enhancement layer**.

The main paper should be:

Geopolitical shocks and European financial stability through volatility connectedness.

The market-implied detector should answer:

Which geopolitical shocks actually become systemic market stress events?

That gives the paper a clean hierarchy:

1. **Main outcome:** European financial fragility index.
2. **Main mechanism:** volatility connectedness.
3. **Enhancement / classification:** market-implied stress regimes.
4. **Supporting evidence:** country-level local projections and event-time volatility.

This keeps the paper coherent and avoids trying to do too much.

10. The best final research question

I would formulate it exactly like this:

Do geopolitical shocks become systemic financial stress events by increasing volatility connectedness across European equity markets, and can market-implied stress regimes distinguish financially material geopolitical shocks from those absorbed by markets?

This combines all three ideas in one coherent paper.